

## Section 5.8

I. Integration  $\int_a^b f(x) dx$

A. If possible – Apply the Fundamental Theorem of Calculus

B. If not possible – Numerical Approximation

1. Partition  $[a, b]$  in  $n$  subintervals

a. 
$$\Delta x = \frac{b-a}{n}$$

b. 
$$x_k = a + k\Delta x$$

2. Approximation by Rectangles

a. Right-hand Rule 
$$I \approx \sum_{k=1}^n f(x_k)\Delta x$$

b. Mid-point Rule

$$(1) \quad x_k^* = \frac{x_k + x_{k-1}}{2} = a + (k - \frac{1}{2})\Delta x$$

$$(2) \quad I \approx \sum_{k=1}^n f(x_k^*)\Delta x$$

3. Trapezoid Rule

a. Area Element 
$$A_k = \frac{1}{2} [f(x_{k-1}) + f(x_k)]\Delta x$$

b. 
$$I \approx \left[ \frac{1}{2} f(x_0) + \sum_{k=1}^{n-1} f(x_k) + \frac{1}{2} f(x_n) \right] \Delta x$$

4. Simpson's Rule

a.  $n$  even

b. Area Element 
$$A_{2k} = \frac{1}{3} [f(x_{2k-2}) + 4f(x_{2k-1}) + f(x_{2k})]\Delta x$$

c.

$$I \approx \frac{1}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 4f(x_{n-1}) + f(x_n)]\Delta x$$

## II. Error Estimates

A. Mid-point Rule  $|E_n| \leq \frac{(b-a)^3}{24n^2} M$  where  $M = \max_{x \in [a,b]} |f''(x)|$

B. Trapezoid Rule  $|E_n| \leq \frac{(b-a)^3}{12n^2} M$  where  $M = \max_{x \in [a,b]} |f''(x)|$

C. Simpson's Rule  $|E_n| \leq \frac{(b-a)^5}{180n^4} K$  where  $K = \max_{x \in [a,b]} |f^{(iv)}(x)|$